

Motus: A Unified Latent Action World Model

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Abstract

While a general embodied agent must function as a unified system, current methods are built on isolated models for understanding, world modeling, and control. This fragmentation prevents unifying multimodal generative capabilities and hinders learning from large-scale, heterogeneous data. In this paper, we propose **Motus**, a unified latent action world model that leverages existing general pretrained models and rich, sharable motion information. **Motus** introduces a Mixture-of-Transformer (MoT) architecture to integrate three experts (i.e., understanding, video generation, and action) and adopts a UniDiffuser-style scheduler to enable flexible switching between different modeling modes (i.e., world models, vision-language-action models, inverse dynamics models, video generation models, and video-action joint prediction models). **Motus** further leverages the optical flow to learn latent actions and adopts a recipe with three-phase training pipeline and six-layer data pyramid, thereby extracting pixel-level “delta action” and enabling large-scale action pretraining. Experiments show that **Motus** achieves superior performance against state-of-the-art methods in both simulation (a +15% improvement over X-VLA and a +45% improvement over $\pi_{0.5}$) and real-world scenarios (improved by +11~48%), demonstrating unified modeling of all functionalities and priors significantly benefits downstream robotic tasks. Project Page: <https://motus-robotics.github.io/motus>.

1. Introduction

A unified model is essential for embodied agents to integrate a spectrum of cognitive functions—from understanding scenes and instructions, imagining possible futures, to predicting consequences and generating actions—into a unified whole. However, existing methods model these

capabilities in isolation: some rely on vision-language-action models (VLAs) [5, 8, 11, 29, 36, 42, 67, 73] to learn static policies from vision and language; others use world models or generative approaches built on predicted futures [4, 7, 22, 24, 28, 32, 45, 48, 60, 63, 70]; and $\mathcal{F}_1$ [37] combines VLAs and inverse dynamics models (IDMs) by explicitly imagining future visual observations, but it excludes world models or video generation models (VGMs), resulting in incomplete unification. These approaches fragment what should be a unified system into 5 separate modeling tasks: VLA, World Model, IDM, Video Generation, and Video-Action Joint Prediction.

Two fundamental challenges (detailed in Sec. 3) hinder the integration of these capabilities. First, *unifying such multimodal generative capabilities* within one framework is nontrivial. While unified world models (UWMs) [72] offer a theoretical prototype, they are typically trained from scratch or with limited priors, lacking either robust vision-language understanding from vision-language models (VLMs) or rich physical interaction knowledge from VGMs. Second, embodied intelligence demands the ability to *learn from large-scale heterogeneous data*—including internet videos, egocentric human demonstrations, and multi-robot trajectories—but action spaces vary widely across embodiments, and most video data lack action labels, making it difficult to pretrain action experts with general motion and interaction priors.

To address these challenges, we propose **Motus**, a unified latent action world model that integrates pretrained experts within a Mixture-of-Transformers (MoT) architecture. Our approach unifies the 5 key distributions by connecting a video generator (generative expert), an action expert, and a vision-language understanding expert via shared multi-head self-attention layers—a design we term **Tri-model Joint Attention**—which preserves specialized functionalities while enabling cross-modal knowledge fusion. To further coordinate multimodal generation, **Motus** incorporates a UniDiffuser-like scheduler, allocating distinct timesteps and noise scales to each modality (e.g., videos and actions).

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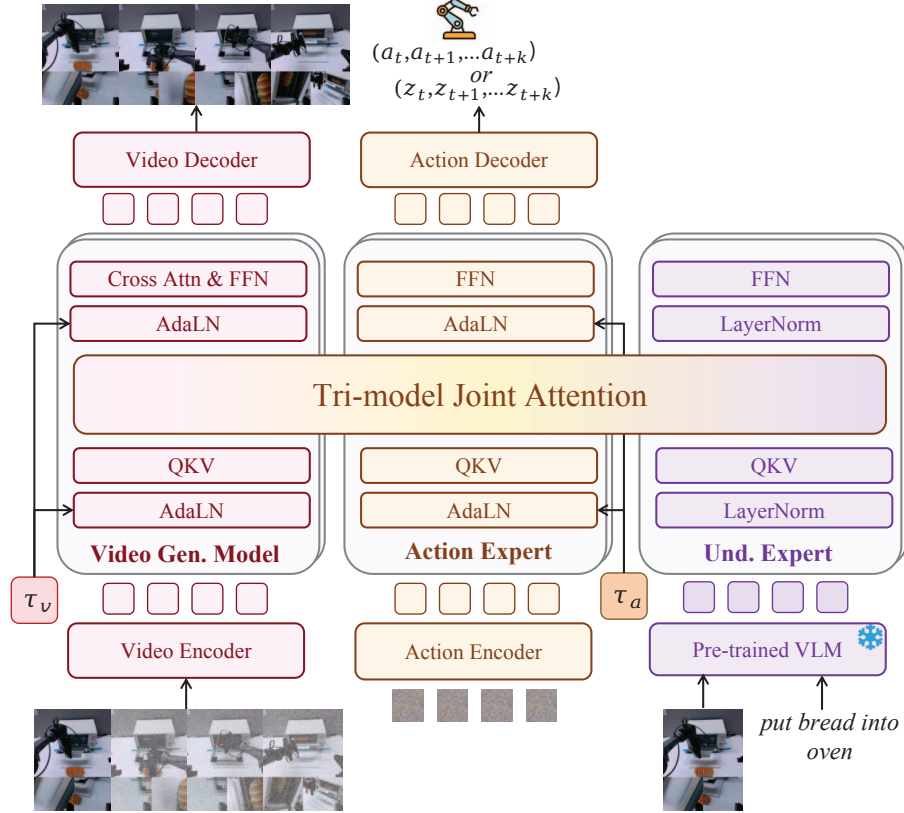


Figure 1. Motus Architecture. Here, $a_t \dots a_{t+k}$ are actions, $z_t \dots z_{t+k}$ are latent actions, and τ_v and τ_a are the rectified flow timesteps for the video generation model and the action expert, respectively.

This enables a unified manner for simultaneous modeling marginal, conditional, and joint distributions, as well as adaptive switching among different inference modes (*e.g.*, VLA, world model, IDM, VGM, Joint Prediction).

Additionally, to leverage heterogeneous data at scale, we introduce latent actions, which encode motion patterns from optical flow as a pixel-level delta action. This representation bridges visual dynamics with control signals, enabling the action expert to be pretrained on diverse unlabeled videos and robot trajectories. Specifically, a pretrained deep compression autoencoder (DC-AE) with additional lightweight downsampling modules is used to reconstruct optical flow, whereas its encoded low-dimensional latents are supervised with a few action labels, both task-related and task-agnostic, thus steering the focus towards patterns associated with robotic activities.

Subsequently, Motus undergoes a three-phase pretraining/finetuning pipeline (*i.e.*, video pretraining, latent action pretraining, and embodiment-specific action finetuning) on a six-layer data pyramid spanning web-scale, egocentric human, simulation, task-agnostic, multi-robotic, and target-robotic data. This recipe aligns behaviors across different embodiments within the motion space described by opti-

cal flows and shares such interaction knowledge with target embodiments to enhance the generalization in downstream tasks, thereby providing the action expert with pretraining like other experts.

Overall, our contributions can be summarized as follows:

- A unified embodied foundation model that integrates five mainstream paradigms (*i.e.*, world models, IDMs, VLAs, VGMs, and video-action joint prediction models) without compromising general multimodal priors.
- A scalable robotic recipe with a three-phase training pipeline and six-layer data pyramid that leverages optical flow-based latent action to learn cross-embodiment transferable motion knowledge.
- Extensive experiments show that Motus significantly outperforms state-of-the-art approaches in both simulation (a **+15%** improvement over X-VLA [67] and a **+45%** improvement over $\pi_{0.5}$ [8]) and real-world scenarios (improved by **+11~48%**), demonstrating that large-scale general and domain-specific priors can be effectively fused to enhance the generalization of policy learning.

2. Related Works

2.1. Unified Multimodal Models

Unified multimodal models jointly model various modalities and tasks within a single generative framework [33, 46, 52, 54, 56, 59], showing broad applications across several domains [41, 61, 71]. In particular, Bagel [20] achieves unification via MoT [35], sharing the multi-head self-attention layers between understanding experts and generation experts. In contrast, existing embodied foundation models are developed independently, spawning multiple disparate paradigms: some leverage the text-image understanding capabilities of VLMs to learn action prediction [6, 8, 30], while others utilize VGMs to generate video sequences and infer actions from consecutive frames [22, 24, 70]. Recently, $\mathcal{F}_1$ [37] extends VLAs to explicitly imagine future visual states and output actions by IDMs, thereby merging both models. Furthermore, UWM [72] unifies world models, VLAs, IDMs, VGMs, and videoaction joint prediction models within a single diffusion backbone, making an initial exploration of complete robotic models. Unlike UWM, our method goes beyond unified modeling by further incorporating internet-scale general multimodal priors and specialized priors from massive robotic trajectories.

2.2. Latent Action Models

Latent actions mitigate the scarcity of action labels by capturing visual dynamics, and are typically derived by coupling IDMs with forward dynamics models (FDMs) to reconstruct the next frame conditioned on the previous one [9, 10, 23, 43]. Initially, RGB images are used for supervision, but this introduces task-irrelevant appearance information [65]. To remove such interference, a common approach is restricting autoencoder’s capacity to encode low-dimensional latents [17, 44, 62], thereby reducing the inclusion of redundancy. AdaWorld [25] attempts to decouple the representations, such as β -VAE [26], in order to retain only the useful factors. Other approaches explore alternative reconstruction objectives, *e.g.*, DINOv2 features [11, 17, 57], object keypoints [19, 58, 64], and language instructions [18], which carries rich semantic and spatial features. Moreover, LAOM [40] employs a few true action labels to encourage them to focus on robotic activities. Building on these advances and inspired by optical flow as a universal motion expression [13, 53, 69], we use it to align cross-embodiment behaviors and learn latent actions to facilitate large-scale pretraining.

3. Problem Formulation and Challenges

Embodied Policies. We consider the task of language-conditioned robotic manipulation. For each embodiment, the task defines an action $\mathbf{a} \in \mathcal{A}$, an observation $\mathbf{o} \in \mathcal{O}$ (visual input), a language instruction $\ell \in \mathcal{L}$, and the propri-

oception of the robot $\mathbf{p}$, where $\mathcal{A}$, $\mathcal{O}$ and $\mathcal{L}$ denote the action space, the observation space, and the language instruction space respectively. The task typically provides an expert dataset $D_{\text{expert}} = \{\{\ell, \mathbf{p}_1, \mathbf{o}_1, \mathbf{a}_1, \dots, \mathbf{p}_N, \mathbf{o}_N, \mathbf{a}_N\}\}$, which contains robot proprioception, visual observations, and actions collected by an expert over N timesteps, along with corresponding language annotations for each trajectory. We train a policy parameterized by θ on D_{expert} . At each timestep t , the policy predicts the next k actions (action chunking [66]) based on the current observation and proprioception, modeling the distribution $p_\theta(\mathbf{a}_{t+1:t+k} \mid \mathbf{o}_t, \mathbf{p}_t, \ell)$ or $p_\theta(\mathbf{a}_{t+1:t+k} \mid \mathbf{o}_t, \ell)$. The policy p_θ is trained to maximize the likelihood objective:

$$\max_{\theta} \mathbb{E}_{(\mathbf{o}_t, \mathbf{p}_t, \mathbf{a}_{t+1:t+k}, \ell) \sim D_{\text{expert}}} \log p_\theta(\mathbf{a}_{t+1:t+k} \mid \mathbf{o}_t, \mathbf{p}_t, \ell). \quad (1)$$

Furthermore, based on the symbolic definitions above, we can derive the probability distributions for the 5 modeling types of embodied intelligence, which can be integrated into a single model for training:

- VLA: $p(\mathbf{a}_{t+1:t+k} \mid \mathbf{o}_t, \ell)$.
- World Model: $p(\mathbf{o}_{t+1:t+k} \mid \mathbf{o}_t, \mathbf{a}_{t+1:t+k})$.
- IDM: $p(\mathbf{a}_{t+1:t+k} \mid \mathbf{o}_{t:t+k})$.
- Video Generation: $p(\mathbf{o}_{t+1:t+k} \mid \mathbf{o}_t, \ell)$.
- Video-Action Joint Prediction: $p(\mathbf{o}_{t+1:t+k}, \mathbf{a}_{t+1:t+k} \mid \mathbf{o}_t, \ell)$.

Challenge 1: Unifying Multimodal Generative Capabilities. A capable embodied agent must integrate a spectrum of cognitive functions—from understanding scenes and instructions, imagining possible futures, to predicting consequences and generating actions—to possess a human-like capacity, as a unified whole. Current models, however, are fragmented and fail to capture the full set of necessary capabilities within one system. This presents a challenge: how to unify the modeling of five key distributions—VLA, World Model, IDM, Video Generation, and Video-Action Joint Prediction—within a single framework. While prior work, such as UWMs [72], has made some progress, a critical limitation persists: these approaches are either trained from scratch, built upon smaller base models, or—even when incorporating some priors—invariably lack the full spectrum of knowledge, missing *either* visual understanding priors from VLMs *or* physical interaction priors from VGMs. Consequently, they lack the comprehensive world knowledge required for robust and generalizable embodied intelligence. Therefore, the nontrivial challenge of jointly modeling various distributions of vision, language, and action within a unified framework remains unaddressed, which is precisely the gap our work fills.

Challenge 2: Utilization of Heterogeneous Data. A central challenge in embodied intelligence is how to make effective use of large scale heterogeneous data. Action spaces vary widely between embodiments in dimension,

range, and semantics, and robots differ in morphology, actuation, and sensing. As a result, control signals are not directly reusable and policies struggle to learn universal priors that transfer across embodiments. Existing approaches, including [8, 31, 36, 50, 67], try to address this by using a general backbone network with embodiment-specific information injection. However, they still depend primarily on labeled robotic trajectories and cannot integrate these datasets with large-scale internet videos or egocentric human videos, which lack action annotations but contain abundant motion and physical interaction cues. This limitation prevents large-scale pretraining of the action expert and reduces the ability to learn general motion priors.

4. Methodology

4.1. Motus

Model Architecture. To address the challenges of unifying multimodal generative capabilities outlined in Sec . 3, we propose Motus, a unified latent action world model. First, Motus is designed as a general generative model that jointly learns on heterogeneous multimodal data, thereby integrating the diverse capabilities (*e.g.*, modeling 5 distributions) of a general-purpose system within a single network. Second, to circumvent the need for impractical amounts of aligned multimodal data, Motus leverages the rich, pre-trained priors of existing foundation models. It integrates a pretrained VGM (generative expert), an understanding expert with pretrained VLM, and an action expert within a Mixture-of-Transformers (MoT) architecture (as shown in Fig. 1), effectively fusing their complementary strengths—encompassing scenes understanding, instructions interpreting, consequences prediction, future video imagination, and action planning—without requiring full end-to-end training from scratch. Unlike Unified World Models (UWMs) [72], which simply concatenate observation tokens and action tokens and process them through a single series of N UWM blocks (containing self-attention and feed-forward network (FFN) layers), our approach leverages pretrained VLMs and VGMs by adopting a MoT structure. In our model, each expert maintains an individual Transformer module, while the multi-head self-attention layers are concatenated, *i.e.*, **Tri-model Joint Attention**. This not only preserves distinct function roles across experts without causing task interference but also enables effective cross-modal feature fusion, encouraging diverse pretrained knowledge to complement one another. During training, Motus jointly predicts chunks of videos and actions with rectified flow-based objectives:

$$l_{\text{action}}^{\theta} = \mathbb{E}_{(\mathbf{o}_{t:t+k}, \mathbf{a}_{t+1:t+k}, \ell) \sim \mathcal{D}} \left\| v_a^{\theta} - (\epsilon_a - \mathbf{a}_{t+1:t+k}) \right\|_2^2, \\ \tau_a \sim \mathcal{U}(0, T_{\tau}) \\ \epsilon_a \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$l_{\text{obs}}^{\theta} = \mathbb{E}_{(\mathbf{o}_{t:t+k}, \mathbf{a}_{t+1:t+k}, \ell) \sim \mathcal{D}} \left\| v_o^{\theta} - (\epsilon_o - \mathbf{o}_{t+1:t+k}) \right\|_2^2, \\ \tau_o \sim \mathcal{U}(0, T_{\tau}) \\ \epsilon_o \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$l^{\theta} = l_{\text{action}}^{\theta} + l_{\text{obs}}^{\theta}.$$

where $\mathbf{o}_t$ is the condition frame, $\mathbf{o}_{t+1:t+k}$, $\mathbf{a}_{t+1:t+k}$ are subsequent observations and actions, τ_a and τ_o are the assigned timesteps, ϵ_a , ϵ_o are the sampled Gaussian noises, v_a^{θ} , v_o^{θ} are velocity field predicted by our unified model, and $l_{\text{action}}^{\theta}$, l_{obs}^{θ} are loss of observations and actions. By allocating different timesteps and noise scales to videos and actions, respectively, Motus establishes a UniDiffuser-like scheduler to capture heterogeneous data distributions and adaptively switch between various embodied foundation models during inference (*e.g.*, VLA, World Model, IDM, VGM, Joint Prediction). The resulting model understands scenes, follows instructions, predicts outcomes, imagines futures, and outputs actions—all within a unified multimodal architecture.

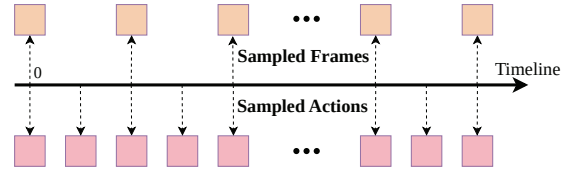


Figure 2. Action-Dense Video-Sparse Prediction. The sampling rates for video frames and actions differ.

Action-Dense Video-Sparse Prediction. Since our model builds upon the widely cited action-chunking technique, Motus needs to predict a chunk of future video and action sequences $\mathbf{o}_{t+1:t+k}$, $\mathbf{a}_{t+1:t+k}$. This leads to several issues: (1) low training and inference efficiency, (2) redundant video frame predictions, and (3) an imbalance in the Tri-modal Joint Attention mechanism—where the number of video tokens significantly exceeds that of action tokens. This imbalance causes the model to overfit to video prediction, thereby weakening its action prediction capability. To address these problems, we propose an Action-Dense Video-Sparse Prediction strategy, as shown in Fig. 2. During both training and inference, we downsample the video frames so that the number of video tokens and action tokens remains balanced—for example, by setting the video frame rate to one-sixth of the action frame rate.

Experts Details. For the generative expert, we employ Wan 2.2 5B [49] as the video foundation model for its accessibility and ease of use. We extend its self-attention context to create a cross-modal Tri-model Joint Attention mechanism. For the action expert, we construct a Transformer block of the same depth as Wan. Each block com-

prises AdaLN for injecting rectified flow timesteps, a Feed-Forward Network (FFN), and the Tri-model Joint Attention for cross-expert interaction. We select Qwen3-VL-2B [2, 3, 51] as the encoder for our understanding expert due to its inherent capabilities in 3D grounding, spatial understanding, and precise object localization, which are crucial for robotic manipulation. The input to this expert is taken from the last-layer corresponding tokens of the VLM. The understanding expert itself consists of several Transformer blocks, each containing Layer Normalization, an FFN, and the Tri-model Joint Attention.

4.2. Latent Actions

We further address Challenge 2 to leverage large-scale heterogeneous data by learning generalizable action patterns directly from visual dynamics. Specifically, we introduce **latent actions** that encode the motion learned directly from pixels. These latent actions allow the model to absorb motion knowledge from various sources such as internet videos, egocentric human demonstrations, and multi-robot trajectories, thereby strengthening the pretraining of action expert even on data without explicit action labels.

Optical Flow Based Representation. We adopt optical flow as a natural representation of motion, which captures pixel-level displacements between consecutive frames. Specifically, optical flows are computed by DPFlow [38] and then converted into RGB images. To compress this high-dimensional representation into a control-level space, we employ a deep convolutional variational autoencoder (DC-AE [14]) that reconstructs the flow while encoding it into four 512-dimensional tokens. A lightweight encoder then projects these concatenated 4×512 features into a 14-dimensional vector, roughly matching the scale of typical robot action spaces. The overall architecture is shown in Figure 3. This dimensional correspondence ensures that the latent representation can align naturally with real robotic controls and act as a bridge between perception and action.

Training and Distribution Alignment. To help align the latent space to realistic action space, we incorporate task-agnostic data following AnyPos [45]. Specifically, task-agnostic data uses Curobo to collect imageaction pairs by randomly sampling the target robots action space in a task-agnostic manner. This data provides additional real action supervision, helping the VAE learn an embedding that reflects feasible motor behaviors and anchors the latent actions to the true control distribution.

During training, we mix 90% unlabeled data for self-supervised reconstruction with 10% labeled trajectories for weak action supervision, where the labeled portion includes both task-agnostic data and standard robot demonstrations. Dimensional correspondence and weak action supervision jointly drive the latent-action distribution to align with the real action distribution, allowing motion priors learned from

videos to naturally map to executable controls.

The total loss combines reconstruction, alignment, and KL regularization:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_a \|a_{\text{real}} - a_{\text{pred}}\|^2 + \beta \mathcal{L}_{\text{KL}}, \quad (2)$$

where $\mathcal{L}_{\text{recon}}$ minimizes flow-reconstruction error, the second term aligns latent and real actions, and $\mathcal{L}_{\text{KL}}$ regularizes the latent space.

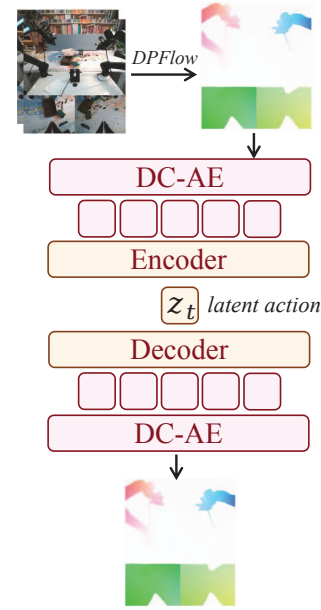


Figure 3. The Latent Action VAE.

4.3. Model Training and Data

Motus Training. Motus is trained in three structured stages (Tab. 1) to progressively integrate physical interaction priors from diverse datasets into a policy transferable to a target robot. Each stage addresses a key challenge:

- **Stage 1: Learning Visual Dynamics.** To anchor the model in realistic physical interactions, we first adapt the Video Generation Model (VGM) using multi-robot trajectories and human videos. This enables the VGM to generate plausible future video sequences of tasks from a language instruction and an initial image.
- **Stage 2: Learning Action Representations.** To bridge visual forecasts with control, we pretrain the entire Motus model (VLM frozen) on videos, language, and latent actions. This stage initializes the action expert by embedding knowledge of motion and interaction into the latent action space.
- **Stage 3: Specializing for the Target Robot.** We finalize the model by fine-tuning it on target-robot data, ensuring that the acquired priors are fully adapted to the specific embodiment’s dynamics and kinematics.

Table 1. **Motus Training.**

Stage	Data	Training
Pretrained Foundation Models (Off-the-shelf)	Level 1: Web Data	VGM and VLM
Stage 1 (Video Generation)	Level 2: Egocentric Human Videos Level 3: Synthetic Data Level 5: Multi-Robot Task Trajectory Data	Only VGM
Stage 2 (Unified Training with Latent Actions)	Level 2: Egocentric Human Videos Level 3: Synthetic Data Level 4: Task-agnostic Data Level 5: Multi-Robot Task Trajectory Data	Motus (all 3 experts, with latent actions)
Stage 3 (SFT)	Level 6: Target-Robot Task Trajectory Data	Motus (all 3 experts, with actions)

Data. To equip robots with generalizable manipulation skills, we leverage large-scale multimodal data that encapsulates rich prior knowledge—from semantic understanding and physical reasoning to spatiotemporal dynamics and decision-making. As outlined in Section 3, embodied data inherently spans multiple modalities: language ℓ , image o , and action a ¹. By considering the presence or absence of each modality, we systematically identify all meaningful data types²:

- **Language + Image + Action:** robot trajectories (e.g., used in VLAs), $\{\ell, o_1, a_1, \dots, o_N, a_N\}$.
- **Language + Image:** video sequences $\{\ell, o_1, \dots, o_N\}$ or image-text pairs $\{(\ell, o)\}$.
- **Image + Action:** task-agnostic interaction data $\{(o_1, a_1, \dots, o_i, a_i)\}$.
- **Language-only:** textual corpora $\{\ell\}$.

We exclude data lacking visual modality (e.g., language + action) as it is unsuitable for visuomotor policy learning. The remaining types form the complete spectrum of useful sources for embodied policy acquisition. To structure this diversity, we introduce the *embodied data pyramid* (Fig. 4), which organizes data types hierarchically by richness and policy relevance.

Our framework effectively integrates and aligns all six data levels—from large-scale but indirect web sources to targeted robot demonstrations—across tailored training stages

¹In joint position control, proprioception and action share the same representation space.

²Language can be annotated post-hoc to support task-oriented learning.

(Tab. 1), unifying heterogeneous datasets [1, 15, 27, 36, 55] within a single, cohesive model architecture.

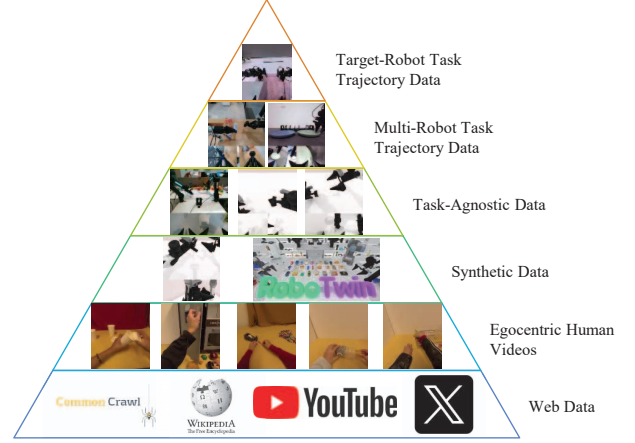


Figure 4. The Embodied Data Pyramid categorizes data into six levels, from Level 1 at the base to Level 6 at the top. Data quantity decreases from bottom to top, while data quality increases. The order of Levels 3 and 4 may sometimes vary.

5. Experiments

We conduct extensive experiments to assess Motus in both simulation and real-world environments.

5.1. Baselines

We compare Motus with the video-action joint prediction mode against several state-of-the-art methods: $\pi_{0.5}$ [8] and X-VLA [67]. We evaluate them in simulation environments and further assess the performance of $\pi_{0.5}$ in real-world tasks. We also compared both the from-scratch and Stage-1-only trained models against our own model.

5.2. Evaluation in Simulation Environment

We evaluated single-task performance on 50 representative manipulation tasks from the RoboTwin 2.0 tasks in randomized scenes. To probe the general ability of our method, we carry out multi-task training: Motus and all baselines are trained on 2500 demonstrations collected in clean scenes (50 per task) plus 25000 demonstrations gathered in heavily randomized scenes (500 per task). The randomization includes random backgrounds, a cluttered table, table-height perturbations, and randomized lighting. All models are fine-tuned for 100K steps on the RoboTwin dataset starting from their pretrained checkpoints, and we evaluate performance by measuring the success rate of each task over 100 execution trials.

As shown in Tab. 2, Motus achieves state-of-the-art performance on the RoboTwin 2.0 randomized multi-task setting, delivering over a 45% absolute improvement compared with the $\pi_{0.5}$ model. By using a unified MoT model,

Table 2. Evaluation on RoboTwin 2.0 Simulation (Clean vs Randomized, 50+ tasks).

Simulation Task	$\pi_{0.5}$		X-VLA		w/o Pretrain		Stage1		Motus	
	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.
<i>Place Dual Shoes</i>	12%	7%	79%	88%	78%	80%	94%	94%	93%	87%
<i>Move Stapler Pad</i>	16%	18%	78%	73%	49%	37%	75%	68%	83%	85%
<i>Stack Blocks Two</i>	48%	56%	92%	87%	96%	94%	99%	99%	100%	98%
<i>Scan Object</i>	42%	38%	14%	36%	42%	50%	56%	69%	67%	66%
<i>Place Object Stand</i>	74%	65%	86%	88%	91%	93%	93%	96%	98%	97%
<i>Place Fan</i>	25%	36%	80%	75%	77%	85%	77%	85%	91%	87%
<i>Move Pillbottle Pad</i>	33%	29%	73%	71%	83%	83%	96%	90%	93%	96%
<i>Pick Dual Bottles</i>	10%	6%	47%	36%	58%	68%	7%	17%	96%	90%
<i>Blocks Ranking Rgb</i>	43%	35%	83%	83%	92%	88%	97%	98%	99%	97%
.....(50 tasks)										
<i>Turn Switch</i>	5%	6%	40%	61%	69%	60%	59%	64%	84%	78%
<i>Pick Diverse Bottles</i>	5%	3%	58%	36%	53%	62%	18%	18%	90%	91%
<i>Place Bread Basket</i>	48%	56%	81%	71%	73%	83%	89%	87%	91%	94%
<i>Stack Blocks Three</i>	15%	16%	6%	10%	71%	76%	99%	95%	91%	95%
<i>Put Bottles Dustbin</i>	12%	9%	74%	77%	36%	33%	34%	24%	81%	79%
<i>Place Can Basket</i>	19%	25%	49%	52%	46%	62%	66%	55%	81%	76%
<i>Stamp Seal</i>	36%	23%	76%	82%	80%	88%	93%	95%	93%	92%
<i>Hanging Mug</i>	3%	3%	23%	27%	14%	10%	37%	25%	38%	38%
<i>Handover Block</i>	18%	19%	73%	37%	34%	15%	55%	55%	86%	73%
<i>Stack Bowls Three</i>	33%	35%	76%	86%	90%	74%	86%	83%	79%	87%
<i>Place Object Basket</i>	43%	36%	44%	39%	74%	75%	76%	80%	81%	87%
<i>Open Microwave</i>	35%	37%	79%	71%	83%	82%	82%	84%	95%	91%
Average (%)	42.98	43.84	72.80	72.84	72.8	77.00	82.86	81.86	88.66	87.02

Motus successfully integrates vision, language, and action generation, solving **Challenge 1**. In **Challenge 2**, the introduction of latent actions enables Motus to effectively leverage both labeled and large-scale unlabeled data, improving generalization across embodiments and capturing rich motion priors. This combination of techniques allows Motus to overcome the limitations of previous approaches and achieve superior performance.

5.3. Real-World Experiments

We evaluate Motus across two distinct real-world dual-arm robotic platforms, Ac-One and Agilex-Aloha-2 under a comprehensive set of non-trivial tasks that span various dimensions of policy capabilities including: (1) Spatial Understanding (2) Deformable Objects Manipulation (3) Precision Fluid Control (4) Visual understanding (5) Long-Horizon Planning, such as fold towel, brew coffee using drip coffee machine and grind coffee beans with grinder.

For each task, we employed 100 trajectories for training. Consistent with the simulator, a multi-task joint training scheme was adopted: all tasks on each robotic platform were trained collectively within a single model, which was subsequently evaluated on every individual task. This approach provides a comprehensive and rigorous assessment of the model’s robustness and generalization capabilities.

Since most tasks involve long-horizon reasoning and are decomposable, we employed the partial success rate for evaluation. This metric quantifies performance by decomposing a task into subtasks, where the model earns par-

tial scores for achieving specific subgoals and a full score only for overall success, thereby offering a more compelling demonstration of its capability. Examples are shown in Tab. 9 and Tab. 10.

The results are reported in Table 3. Our results demonstrate that Motus significantly outperforms the baseline $\pi_{0.5}$ across all tasks on both robotic arms. Visualizations are provided in Figure 5.

5.4. Ablation Study

All ablation experiments are conducted on 27.5K RoboTwin simulation data and trained from scratch by 50K steps.

Training Stages. We analyze the contribution of each training stage. This involves benchmarking models without pretraining, only stage 1 pretraining, and both stage 1&2 pretraining. Evaluations are carried out in the RoboTwin 2.0 simulator to measure accuracy. In real-world deployments we compare Motus against its from-scratch counterpart. The results in simulator are summarized in Fig 6, and results in real-world experiments are shown in Table 3.

Expert Components. As shown in Tab. 4, both VLM and VGM are critical to Motus, with VGM playing a more dominant role. Specifically, removing VGM leads to a drastic drop in success rate from 77% to 25.5%. In contrast, removing VLM results in a moderate degradation to 64.94%.

Diffusion Schedulers. Under the video-action joint prediction mode, we compare the performance of UniDiffuser-style asynchronous against Joint Diffuser-style synchronous

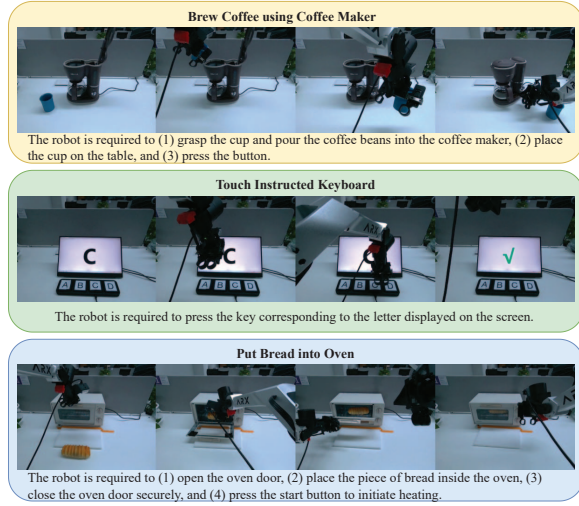


Figure 5. **Task Definitions and Visualizations.** For each task, we describe its language instruction and definitions of each sub-task.

Table 3. Robotic Manipulation Tasks Performance Across Platforms (Partial Success Rate %).

Task Description	$\pi_{0.5}$	w/o Pretrain	Motus
AC-One			
Fold Towel	4	1	14.5
Brew Coffee using Coffee Maker	0	0	62
Get Water from Water Dispenser	30	8	36
Place Cube into Plate	46	60	100
Place Cube into Plate(OOD)	28.125	18.75	75
Grind Coffee Beans with Grinder	8	0	92
Pour Water from Kettle to Flowers	5	5	65
Touch Instructed Keyboard	0	100	82.5
Put Bread into Oven	12	40	42
Average	14.79	25.86	63.22
Agilex-Aloha-2			
Fold Towel	27.5	0	39
Get Water from Water Dispenser	62	8	96
Pour Water from Kettle to Flowers	45	40	47.5
Touch Instructed Keyboard	72.5	85	80
Put Bread into Oven	36	0	34
Average	48.60	26.60	59.30

Table 4. Ablation study on Motus Expert components.

Variant	Success Rate (%)
w/o VLM & Und Expert	64.94
w/o VGM	25.50
Motus (full)	77.00

timestep allocation across modalities. As shown in Tab. 5, UniDiffuser achieves a +9.79% gain in success rate, and Joint Diffuser lacks the flexibility to evolve into other modes, *e.g.*, world model or IDM.

Scaling Law. As shown in Fig. 7, we demonstrate world models exhibit superior scalability and generalizabil-

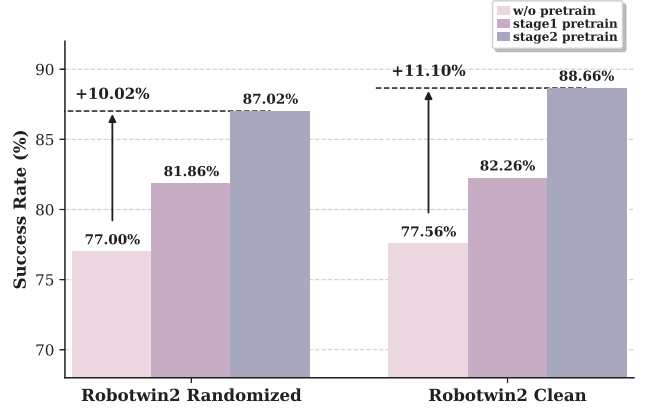


Figure 6. **Ablation in RoboTwin 2.0 Randomized Multi-task Setting.** The figure presents the total success rates (%) of the original Motus (Stage 2 Pretrain) and its two variants: Without Pretrain and Stage 1 Pretrain.

Table 5. Ablation Study on Diffusion Schedulers.

Method	Success Rate (%)
Joint Diffuser	67.21
UniDiffuser (ours)	77.00

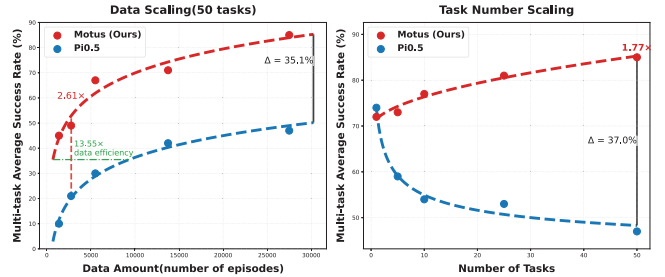


Figure 7. Scaling Curve Comparison between Motus and $\pi_{0.5}$.

ity compared to VLAs across both increasing task diversity and data size. Specifically, as task number increases, $\pi_{0.5}$ degrades while Motus improves with 1.77 $\times$ higher success rate. Under the same data budget, Motus reaches up to 2.61 $\times$ higher success rate, and achieves the same performance with 13.55 $\times$ data efficiency.

6. Conclusion and Limitations

In this work, we present Motus, a unified latent action world model that integrates five mainstream embodied paradigms, and further leverages optical flow-based latent actions to capture embodiment-agnostic motion priors and unlock action pretraining. Extensive experiments across simulation and real-world environments demonstrate that Motus consistently outperforms existing state-of-the-art methods. We hope Motus can inspire future research on world action models (WAMs) and video-action models (VAs), and deeper VLM integration and latent actions learned from internet-scale universal videos can be further involved.

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